

If $y = f(x)$ is linear,
then $\Delta y = m \cdot \Delta x$;
the constant m is
rate, slope, and multiplier
for all x .

If $y = f(x)$ is locally linear,
then $\Delta y \approx f'(a) \cdot \Delta x$;
the derivative $f'(a)$ is
rate, slope, and multiplier
for x near a .